

Week 2

The Board Number Is Wrong

Reconcile disagreeing dashboards and build gates that stop the disagreement returning.

BUILD

BREAK

DEFEND

Why this week matters

The practical motivation

- A model trained on untrusted data only makes bad numbers faster.
- Agents are excellent at writing joins; they are also excellent at hiding missing records.
- Reconciliation is where analytics credibility is won.

Finance says churn is 11.8 percent. Product says it is 9.4 percent. Both numbers are already in draft board papers.

The studio rhythm

1

BUILD

Use the agent to create the first useful version.

2

BREAK

Attack the draft with bad data, edge cases, and misuse.

3

DEFEND

Keep only what survives evidence, tests, and explanation.

Mission brief

Find the trusted number, explain the difference, and build a data-quality gate.

You will build

Reconciliation pipeline with source counts, join coverage, quarantine rows, and validation tests.

Success looks like

- The artefact runs from a clean checkout.
- At least one agent failure is found and fixed.
- The final claim has evidence a peer can inspect.

Start here

Use these files and assumptions before asking the agent to build.

Starter materials

- Starter_Kit/sample_data/week2_subscriptions.csv
- Starter_Kit/sample_data/week2_cancellations.csv
- Starter_Kit/sample_data/week2_product_events.csv
- Student_Workbook.md Week 2 checklist

Confirm first

- Which file owns billing status?
- Which file owns product activity?
- How will unmatched rows be quarantined?

Worked example to imitate

Use Worked_Examples.md and Starter_Kit/examples/ for the full artefact.

What strong evidence looks like

- Good row-count bridge: reports source counts before every join.
- Good quarantine row: C204 is kept, flagged, and explained.
- Good board note: separates billing churn from product activity.

Discuss before building

- What makes this example inspectable?
- Which claim is supported by evidence?
- What limit is still named?

Concept burst

Only the ideas needed for today's build

1

Source counts before transformation.

2

Join coverage and quarantine tables.

3

Validation tests as business controls.

Guided practice menu

Use the matching file in Exercises/ for timing, instructions, and instructor notes.

Core exercises

- Source profile: count rows and keys before joining.
- Join debate: compare inner, left, and full outer review.
- Quarantine table: explain records that should not disappear.

Submit by the end

- One concrete artefact.
- One reproduced or plausible failure.
- One evidence-linked defence note.

Three-hour delivery path

Use these slides as pacing support: set context, practise, build, break, repair, defend.

Week 2

180-minute run of show

- 1 0-5: retrieval link and setup check.
- 2 5-15: cold open and stakeholder pressure.
- 3 15-30: demo target and success criteria.
- 4 30-50: concept burst with worked example.
- 5 50-80: guided practice and prompt clinic.
- 6 80-115: Sprint A builds the first version.
- 7 115-135: failure cards and defect exposure.
- 8 135-165: Sprint B repairs and evidence pack.
- 9 165-180: peer demo, defence note, exit ticket.

Learning outcomes for today

Build

Create the first version of: Reconciliation pipeline with source counts, join coverage, quarantine rows, and validation tests.

Break

Expose at least one agent, data, prompt, model, or workflow failure.

Defend

Explain what can be trusted, what was rejected, and what still needs work.

Team roles

Driver

Operates the repo, files, agent, notebook, tool, or spreadsheet.

Reviewer

Reads outputs, checks assumptions, challenges claims, and watches for defects.

Reporter

Maintains the evidence pack, AI-use log, and defence note.

Retrieval link

Ask the room

- What did we build, break, and defend last week?
- Which habit matters again today?
- Which previous artefact becomes input for this mission?

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Cold open discussion

Ask the room

- Finance says churn is 11.8 percent. Product says it is 9.4 percent. Both numbers are already in draft board papers.
- What would go wrong if the team trusted the agent too early?
- What evidence would make the stakeholder comfortable?

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Stakeholder lens

1 Name the stakeholder and the decision they must make.

2 Name the artefact they expect to receive.

3 Name the risk if the artefact is wrong.

4 Name the evidence that would reduce that risk.

5 Name what should be out of scope today.

What makes this an AI-agent course?

Delegation

Students delegate bounded work to the agent instead of doing everything manually.

Supervision

Students inspect, test, reject, and repair agent output.

Accountability

Students defend the final artefact as human analysts, not as agent operators.

Understand the problem before prompting

Students should know the files, definitions, stakeholder, and acceptance criteria before the agent starts.

Starter file walkthrough

- Starter_Kit/sample_data/week2_subscriptions.csv
- Starter_Kit/sample_data/week2_cancellations.csv
- Starter_Kit/sample_data/week2_product_events.csv
- Student_Workbook.md Week 2 checklist

Confirm before prompting

Ask the room

- Which file owns billing status?
- Which file owns product activity?
- How will unmatched rows be quarantined?

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Good artefact, weak artefact

Good

Runs or can be inspected, names assumptions, and includes checks.

Weak

Looks polished but hides definitions, dropped records, leakage, or unsupported claims.

Repairable

Imperfect first draft with visible evidence and a clear next fix.

Control the agent before it controls the task

Students practise turning vague requests into supervised work orders.

Week 2

Weak prompt pattern

Too broad

The agent chooses the task boundary and success criteria.

Too trusting

The agent chooses definitions, joins, thresholds, or causal claims.

Too vague

The output cannot be checked because no acceptance criteria were given.

Strong work-order pattern

- 1 Business goal: what decision or workflow does this support?
- 2 Inputs: files, tables, definitions, sources, or prior artefacts.
- 3 Task: the bounded work the agent should perform.
- 4 Acceptance criteria: what must be true before humans accept it.
- 5 Constraints: what the agent must not invent, drop, infer, or overclaim.
- 6 Output format: code, table, memo, test, checklist, or tool.

Activity 1 - prompt rewrite

10 min

Source profile: count rows and keys before joining.

Output before moving on

- One revised work order.
- One acceptance criterion.
- One constraint that prevents a known failure.

Prompt rewrite debrief

Debrief prompts

- Which word in the prompt reduced ambiguity?
- Which acceptance criterion can actually fail?
- Which constraint protects the stakeholder?

Instructor listens for

- A specific artefact, not a general intention.
- A named failure mode.
- Evidence that can be inspected.
- A limit or no-use condition.

Concept Depth

Short teaching, immediate use

Use these concept slides to equip the build, not to turn the seminar into a lecture.

Week 2

Concept 1: Source counts before transformation.

Meaning

How this idea changes today's artefact:
Reconciliation pipeline with source counts, join coverage, quarantine rows, and validation tests.

Failure prevented

Connect it to one failure card: Silent inner join removes customers without product events.

Evidence

Students must show:
Reconciliation report with both old numbers and the corrected number.

Concept 1 check

Ask the room

- What would the agent do if this concept were ignored?
- What would a strong student artefact include?
- What is the smallest evidence item that proves the concept was used?

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Concept 2: Join coverage and quarantine tables.

Meaning

How this idea changes today's artefact:
Reconciliation pipeline with source counts, join coverage, quarantine rows, and validation tests.

Failure prevented

Connect it to one failure card: Cancellation date and billing date define churn differently.

Evidence

Students must show:
Quarantine table and row-count bridge.

Concept 2 check

Ask the room

- What would the agent do if this concept were ignored?
- What would a strong student artefact include?
- What is the smallest evidence item that proves the concept was used?

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Concept 3: Validation tests as business controls.

Meaning

How this idea changes today's artefact:
Reconciliation pipeline with source counts, join coverage, quarantine rows, and validation tests.

Failure prevented

Connect it to one failure card: Null reason codes are imputed as 'unknown' and treated as valid.

Evidence

Students must show:
Validation tests that fail on the bad join.

Concept 3 check

Ask the room

- What would the agent do if this concept were ignored?
- What would a strong student artefact include?
- What is the smallest evidence item that proves the concept was used?

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Worked Example

Model the evidence standard

Students should imitate the structure of evidence, not copy the example.

Week 2

Worked example highlights

- Good row-count bridge: reports source counts before every join.
- Good quarantine row: C204 is kept, flagged, and explained.
- Good board note: separates billing churn from product activity.

Worked example critique

Ask the room

- What claim is being made?
- Where is the evidence?
- What limit is named?
- What would you ask before accepting it?

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Make students do the thinking before the build

The goal is not coverage. The goal is a better first sprint.

Week 2

Exercise 1

12 min

Source profile: count rows and keys before joining.

Output before moving on

- A visible table, prompt, test, note, or decision.
- One assumption written down.
- One peer or instructor question answered.

Exercise 1 debrief

Debrief prompts

- What was easy for the agent?
- What decision had to stay human?
- What evidence changed the team's mind?

Instructor listens for

- A specific artefact, not a general intention.
- A named failure mode.
- Evidence that can be inspected.
- A limit or no-use condition.

Exercise 2

12 min

Join debate: compare inner, left, and full outer review.

Output before moving on

- A visible table, prompt, test, note, or decision.
- One assumption written down.
- One peer or instructor question answered.

Exercise 2 debrief

Debrief prompts

- What was easy for the agent?
- What decision had to stay human?
- What evidence changed the team's mind?

Instructor listens for

- A specific artefact, not a general intention.
- A named failure mode.
- Evidence that can be inspected.
- A limit or no-use condition.

Exercise 3

18 min

Quarantine table: explain records that should not disappear.

Output before moving on

- A visible table, prompt, test, note, or decision.
- One assumption written down.
- One peer or instructor question answered.

Exercise 3 debrief

Debrief prompts

- What was easy for the agent?
- What decision had to stay human?
- What evidence changed the team's mind?

Instructor listens for

- A specific artefact, not a general intention.
- A named failure mode.
- Evidence that can be inspected.
- A limit or no-use condition.

Sprint A

Build the first version

Time-box the build so students have something real to attack.

Week 2

Sprint A operating rules

- 1 Start from the work order, not a blank chat.
- 2 Keep the agent output visible and reviewable.
- 3 Run the artefact before improving it.
- 4 Write down the first defect or uncertainty.
- 5 Commit or save the first version before repair.

Sprint A task list

- Ask the agent to profile the source tables.
- Create a reconciliation notebook or script.
- Produce source counts and first churn calculation.

Sprint A checkpoint

Ask the room

- What runs or can be inspected right now?
- What did the agent assume?
- What evidence is missing?
- What is the first defect to investigate?

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Instructor roaming script

- 1 Ask: show me the artefact.
- 2 Ask: show me what the agent changed.
- 3 Ask: show me one check or known value.
- 4 Ask: what would make this unsafe?
- 5 Do not solve unless the team is blocked by setup rather than thinking.

Break Moment

Failure is the teaching device

The class now attacks the first draft so students learn what not to trust.

Week 2

Failure card 1

Defect

Silent inner join
removes customers
without product events.

How to expose it

Create a check,
adversarial example,
missing-value case,
source comparison, or
peer question.

What to record

Save the failing output
before repairing it. This
becomes portfolio
evidence.

Failure card 2

Defect

Cancellation date and billing date define churn differently.

How to expose it

Create a check, adversarial example, missing-value case, source comparison, or peer question.

What to record

Save the failing output before repairing it. This becomes portfolio evidence.

Failure card 3

Defect

Null reason codes are imputed as 'unknown' and treated as valid.

How to expose it

Create a check, adversarial example, missing-value case, source comparison, or peer question.

What to record

Save the failing output before repairing it. This becomes portfolio evidence.

Defect wall

Ask the room

- Post one failure your team found.
- Mark whether it was agent, data, prompt, model, tool, or claim failure.
- Write the evidence that exposed it.

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Sprint B

Repair, harden, and produce evidence

Students now move from one-off fixing to recurrence prevention.

Week 2

Repair moves

- Add join-coverage tests.
- Create a quarantine table for unmatched and ambiguous records.
- Write the business definition beside the code.

Evidence design checklist

- 1 Name the claim the artefact makes.
- 2 Name the failure that could make the claim false.
- 3 Show the check, test, trace, source, or review that exposes the failure.
- 4 Show what changed after repair.
- 5 State the remaining limit.

Sprint B checkpoint

Ask the room

- Can you reproduce the failure?
- Can you show the fix?
- Would the failure be caught earlier next time?
- What will you put in the evidence pack?

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Make evidence social

A peer should be able to inspect the artefact and challenge the defence.

Week 2

Peer demo protocol

- 1 One minute: run or show the artefact.
- 2 One minute: show the failure found.
- 3 One minute: show the repair evidence.
- 4 One minute: peer asks what still should not be trusted.
- 5 One minute: team writes one improvement before submission.

Peer review questions

Artefact

What did you build, and can someone else inspect it?

Failure

What did the agent, data, model, tool, or answer get wrong?

Defence

Which claim is supported, and what remains out of scope?

Turn activity into assessable evidence

The defence note is where students show judgement, not just effort.

Week 2

Defence note writing

10 min

The number I would put in the board paper is...

Output before moving on

- One claim the team can defend.
- One evidence item linked to that claim.
- One remaining risk or no-use condition.

Evidence pack requirements

- Reconciliation report with both old numbers and the corrected number.
- Quarantine table and row-count bridge.
- Validation tests that fail on the bad join.

Rubric lens

Useful artefact

It runs, can be inspected, or clearly solves the mission within scope.

Agent supervision

The team shows what was accepted, rejected, changed, and verified.

Defensible judgement

Claims are specific, evidence-linked, and include limits.

Close

Make the habit portable

Students should leave with one artefact, one failure, and one defensible claim.

Week 2

Common traps to avoid

- Calling the biggest table the source of truth.
- Dropping rows to make the pipeline pass.
- Mixing business reconciliation with data cleaning without labelling which is which.

Stretch options for fast teams

- Build a tiny data-quality dashboard.
- Add a daily alert when join coverage drops below threshold.

After-class handoff

- 1 Commit or save the artefact.
- 2 Save the failed or rejected output.
- 3 Complete the AI-use log.
- 4 Submit the defence note.
- 5 Bring one unresolved question to the next retrieval link.

Agent work order

Give the agent a job, not a wish.

Use this structure

- Goal: reconcile Finance and Product churn numbers.
- Inputs: subscriptions.csv, cancellations.csv, product_events.csv.
- Acceptance: source counts reported, unmatched rows quarantined, churn formula documented.
- Do not drop records silently. Every lost row needs a reason.

Studio sprint A - build the first version

Tasks

- Ask the agent to profile the source tables.
- Create a reconciliation notebook or script.
- Produce source counts and first churn calculation.

Instructor watchlist

- Is the prompt specific enough?
- Did the team run the artefact?
- Did the team record what the agent changed?

Failure cards

Release these after Sprint A. Students must expose the defect before fixing it.

Card 1

**Silent inner join
removes customers
without product
events.**

Card 2

**Cancellation date
and billing date
define churn
differently.**

Card 3

**Null reason codes
are imputed as
'unknown' and
treated as valid.**

Studio sprint B - repair and harden

Repair moves

- Add join-coverage tests.
- Create a quarantine table for unmatched and ambiguous records.
- Write the business definition beside the code.

Definition of done

- The failure is reproducible.
- The fix is committed.
- A future failure would be caught earlier.

Evidence pack

What students should submit

- Reconciliation report with both old numbers and the corrected number.
- Quarantine table and row-count bridge.
- Validation tests that fail on the bad join.

Peer demo

Demo in three minutes

- Run the artefact.
- Show the failure you found.
- Show the evidence that supports the final claim.

Peer questions

- What did the agent get wrong?
- What would fail next?
- What should not be used yet?

Common traps

Watch for these

- Calling the biggest table the source of truth.
- Dropping rows to make the pipeline pass.
- Mixing business reconciliation with data cleaning without labelling which is which.

If your team finishes early

Stretch options

- Build a tiny data-quality dashboard.
- Add a daily alert when join coverage drops below threshold.

Exit ticket

The number I would put in the board paper is...

Write one sentence. It goes into your portfolio evidence.

Leave with this

The trusted number is the one with a row-count story, not the one with the nicer chart.

BUILD / BREAK / DEFEND